DriveCompose: Compositional Generative Reconstruction of Driving Scenes via Joint Completion

Anonymous CVPR submission

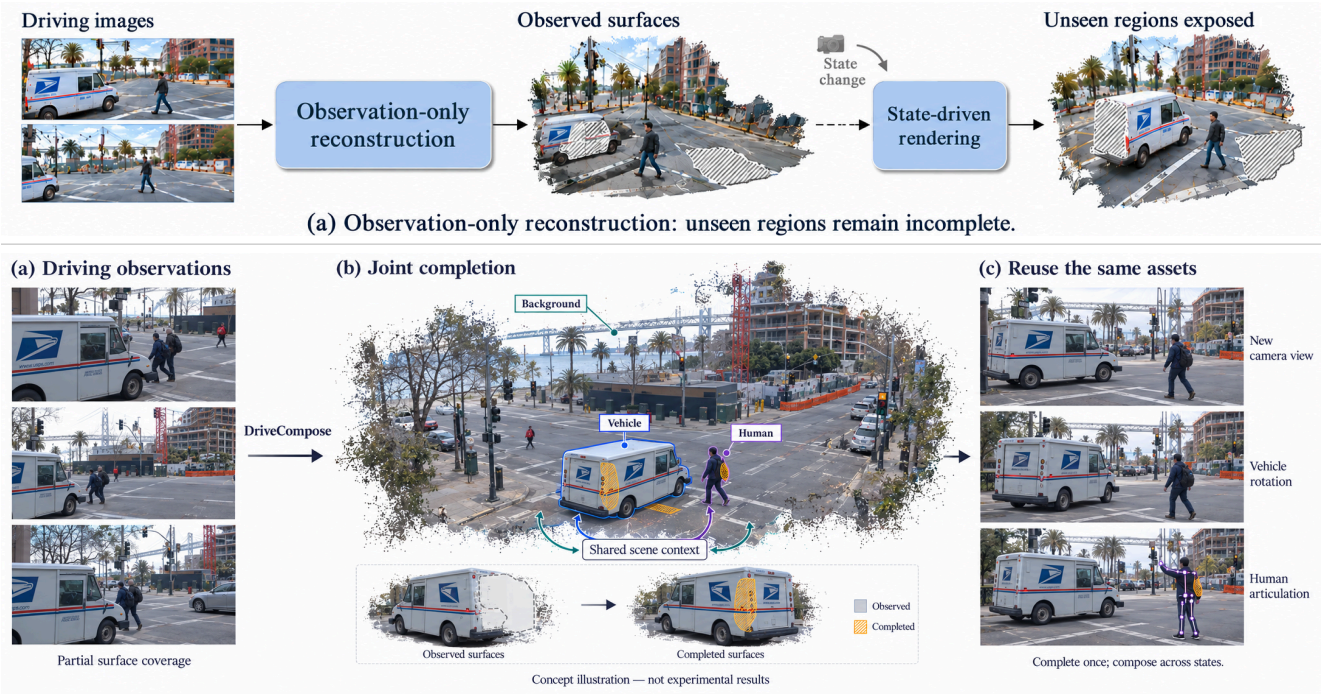

Figure 1. **From observations to reusable completed assets.** Top: viewpoint and actor changes expose missing surfaces in observation-only reconstruction. Bottom: structured encoding and joint completion produce background, canonical vehicles, and humans for reuse across states. Gold indicates completed regions. All scene and asset images are schematic illustrations.

Abstract

Moving the camera or changing actor poses exposes surfaces that were never observed in the input driving images. Reconstructing scenes for simulation and editing therefore requires completing these surfaces while retaining control over actor motion. We present DriveCompose, a framework for compositional generative reconstruction from calibrated driving image sequences. Visual evidence is aggregated in world coordinates for the background and canonical frames for vehicles and humans. A conditional flow model uses cross-component memory to share scene context and jointly complete missing geometry and appearance in the component latents. The completed latents are decoded into Gaussian components with rigid and skeletal bindings, allowing generated surfaces to move with their corresponding actors. The same background and actor assets can then be assembled and rendered under new camera viewpoints and actor states. Experiments on Waymo show higher reconstruction accuracy and better off-trajectory rendering quality than feed-forward reconstruction baselines. Actor editing experiments further demonstrate that completed vehicles and humans can be reused across motion and pose changes.

1. Introduction

Driving simulation and scene editing require 3D scenes that support changes in camera viewpoint, vehicle motion, and human pose [46, 49]. However, driving images reveal only the surfaces visible from the recorded viewpoints and actor configurations [1, 31]. Moving a camera, turning a vehicle, or changing a pedestrian's pose exposes previously unobserved regions (Fig. 1). Reconstructing scenes for these op-

erations therefore requires completing the missing geometry and appearance while ensuring that completed vehicle and human surfaces move with their respective actors.

Recent feed-forward methods fuse multiple images to directly predict driving-scene geometry and Gaussian representations [3, 6, 45, 48]. Motion and visibility estimates align dynamic objects and aggregate surfaces visible across the input sequence. Surfaces absent from the entire sequence call for generative priors over geometry and appearance.

Generative reconstruction uses asset priors, Gaussian completion, or generated views to infer unobserved content [17, 44, 47, 51]. Compositional reconstruction represents the background and moving actors separately to support vehicle motion and human pose control [5, 43, 49]. Completing these component representations lets generated surfaces retain their actors' motion bindings. The visible background and other actors provide scene context for completing a partially observed component.

We introduce DriveCompose, which jointly completes the background, vehicles, and humans as reusable Gaussian components (Fig. 2). The background is represented in world coordinates, rigid assets in their own frames, and human surfaces in a reference pose with skeletal bindings [5, 43]. Cross-component memory shares these components' features to condition flow-based completion [18]. Gaussian decoders preserve their ownership and bindings [15], allowing completed surfaces to move with their actors under new camera and actor states (Secs. 3.1 and 3.4).

Experiments on Waymo show higher reconstruction accuracy and better off-trajectory rendering quality than the feed-forward baselines. Paired synthetic tests compare joint completion with scene-level generation and cross-object coupling, while real actor edits demonstrate reuse of the same completed assets (Sec. 4).

Our contributions are:

- We introduce DriveCompose to jointly complete backgrounds, vehicles, and humans from image sequences as compositional Gaussian assets, reusable under camera, rigid, and skeletal state changes.
- We develop a cross-component latent completion model based on conditional flow matching. Local attention models spatial relationships within each component, while cross-component memory aggregates background and actor features to provide scene context for generating each component's missing content.
- We establish the contribution of shared scene context through matched completion mechanisms and visibility-stratified evaluations, and demonstrate state-driven reuse with real actor edits.

2. Related work

Driving-scene reconstruction. Urban reconstruction builds on NeRF and Gaussian radiance representations [15, 22], with driving-specific rendering developed in NeuRAD and SplatAD [11, 33]. Feed-forward methods predict scene geometry and Gaussians directly from images [2, 4, 12, 35, 36]. STORM aggregates dynamic Gaussians through scene flow, while DGGT combines camera recovery with visibility and motion-guided interpolation [3, 45]. Surround-view inputs, temporal processing, point-aligned representations, and simulation-oriented reconstruction are explored by DrivingForward, StreetForward, PointForward, and Instant NuRec [6, 14, 32, 48]. TokenGS uses learnable tokens to decouple Gaussian prediction from pixels [29].

Compositional scene representations. Separating background and moving components supports actor control in Neural Scene Graphs, Street Gaussians, and DrivingGaussian [23, 43, 50]. OmniRe represents rigid and articulated actors in canonical frames, while UniSim and HUGSIM connect reconstructed content to closed-loop simulation [5, 46, 49]. At the asset level, SMPL-based body models and animatable Gaussian methods provide canonical surfaces and motion bindings [13, 16, 20, 25, 27]; LHM and IDOL infer animatable humans, and UrbanCAD targets controllable vehicles [21, 28, 52]. These frames and bindings provide the interface for reusing surfaces after reconstruction or completion.

Generative reconstruction and completion. Generative reconstruction infers unobserved geometry and appearance through generated views [9, 19, 30] or structured latent representations [41, 42]. GenAssets learns priors for driving assets, CompleteSplat generates hidden Gaussian surfaces, and JRM couples object latents to aggregate unaligned observations [17, 40, 47]. LatentWorld organizes semantic occupancy generation around entity and background latents [24]. DriveCompose jointly completes world-space background and canonical rigid and human codes, retaining their geometry, appearance, and motion bindings for state-driven rendering. Our matched comparisons adapt scene-level generation and cross-object coupling to the same reconstruction interface (Sec. 4.5).

Driving-scene generation. MagicDrive, DriveDreamer, and Vista generate driving imagery conditioned on scene structure or control [8, 10, 37]. StreetCrafter uses generated views to supervise dynamic Gaussian reconstruction, and HorizonForge combines scene editing with video diffusion [38, 44]. WorldSplat generates Gaussians and refines

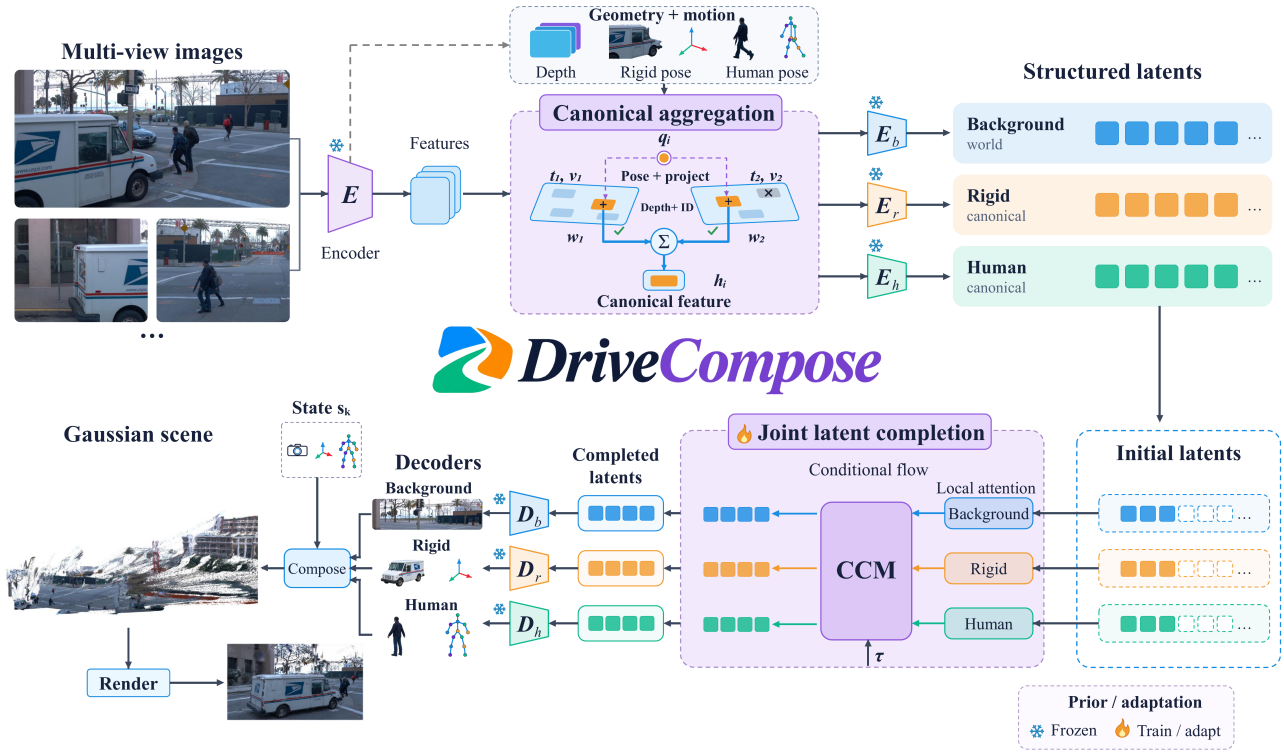

Figure 2. **DriveCompose overview.** Visual encoding and canonical aggregation recover component features; type-specific encoders, joint completion, and Gaussian decoders produce background, vehicle, and human content for state-driven composition. Scene and asset images are schematic illustrations.

their rendered videos, while DIFIX3D+ applies diffusion-based image refinement [39, 51]. DriveCompose stores the completed background and canonical assets, then renders the same 3D content under new camera and actor states.

3. Method

DriveCompose aggregates observations in world and canonical coordinates, jointly completes their latents, and decodes reusable Gaussian assets for rendering under new states (Fig. 2).

3.1. Problem formulation

The input is a synchronized image log $\mathbf{I}_{t,v}$ with timestamps, camera intrinsics, and camera-to-rig calibration. The controlled evaluations provide rig poses, instance identities, vehicle boxes and root trajectories, and target human joints. Human shape and binding are recovered from context; no target RGB or test LiDAR enters reconstruction. Coordinate conventions and image preprocessing are detailed in Appendix D of the supplementary material.

The output is a compositional Gaussian scene

$$\mathcal{W} = \big(\mathcal{G}_b, C_{\text{sky}}, V_{\text{sky}}, \{\mathcal{A}_a\}, \mathcal{B}, \xi\big), \tag{1}$$

where $\mathcal{G}_b$ is the world-space background, C_{sky} a sky cube-map, V_{sky} its supported directions, $\mathcal{A}_a$ the canonical asset for identity a, $\mathcal{B}$ the finite spatial support, and ξ the sampling seed. The components are rendered by Gaussian splatting [15]. Vehicles are represented as rigid assets with fixed local geometry and dimensions. Humans additionally retain a rest skeleton and skinning weights. Scene state controls the placement and articulation of these fixed components.

We model missing content through conditional generation. Let $\mathcal{Q}$ be the recovered query layout and C its observed evidence. The completion model samples unknown codes $\mathbf{z}_u$, while the codec D_ψ decodes the assembled codes into scene content:

$$\mathbf{z}_u \sim p_\theta(\mathbf{z}_u \mid C, \mathcal{Q}), \qquad \mathcal{W} = D_\psi(\mathbf{z}^{\text{obs}}, \mathbf{z}_u; \mathcal{Q}). \tag{2}$$

The query layout assigns each latent code to a spatial location and scene component. Completion fills the unknown codes within this layout, and decoding preserves their component identities and motion bindings.

3.2. Visual recovery and canonical aggregation

The Driving Log Encoder extracts patch features from each image and exchanges pooled context across cameras and

timestamps [7, 34]. Geometry heads predict camera-axis depth, normals, confidence, semantics, and instance assignments, linking image features to 3D surfaces and their owning components. Trajectory-level identity slots separate persistent identity, dimensions, and human shape from frame-dependent root poses and joint rotations. Architectural details and state parameterization are given in Appendix D of the supplementary material.

Spatial supports. A query stores canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local attention group g_i, and reference-world coordinate $\mathbf{q}_i^w$. Reliable depth samples provide surface anchors. Background anchors are combined with a finite world grid; rigid anchors are aligned by inverse SE(3) transforms and combined with a grid in the local bounding volume. Human queries lie on the canonical template surface. Sky queries are unit directions grouped by cubemap face. These supports provide explicit locations for both observed and missing content.

The layout retains canonical coordinates to identify component surfaces and first-input world coordinates $\mathbf{q}_i^w$, including human skinning, to locate them relative to the background and other actors.

Motion-aware evidence aggregation. To aggregate evidence for the same surface across time, we place each canonical query at the corresponding actor state:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \text{xyz}(\mathbf{T}_a^w(t)\bar{\mathbf{q}}_i), & k_i = \text{rigid}, \\ \text{xyz}(\mathbf{T}_h^w(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{3}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Projection gives query depth $z_{i,t,v}$ and samples image features $\boldsymbol{\phi}_{i,t,v}$, depth $d_{i,t,v}$, and confidence $c_{i,t,v}$. We weight observations by confidence and normalized depth residual $r = (d - z)/(\eta_{\text{abs}} + \eta_{\text{rel}} d)$. The indicator $a_{i,t,v}$ selects visible, depth-consistent observations of the same instance (Appendix D):

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r_{i,t,v}^2/2), \tag{4}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v}[\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{5}$$

The masks m_i, f_i, and u_i distinguish observed surfaces, ray-labeled free space, and unknown content, with $m_i + f_i + u_i = 1$. Confidence is the maximum over valid observations; two reliable depth-ray votes establish a free-space label. Our main protocol uses center-ray voting (C); the footprint alternative (F) tests the projected cell extent. Sky features are aggregated by viewing direction using camera rotation and sky semantics. Appendix D reports both rules and their downstream effects.

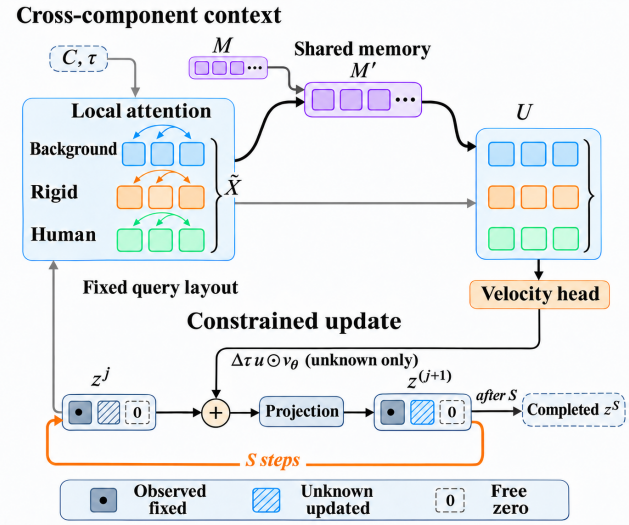

Figure 3. **Joint latent completion.** CCM reads local features and writes back shared context; projected flow updates only unknown codes.

3.3. Joint latent completion

Latent encoding. Type-specific encoders map the 129-dimensional aggregate and its spatial support, $[\mathbf{h}_i; \mathbf{q}_i; \boldsymbol{\delta}_i]$, through a $135 \to 128 \to 128$ MLP with LayerNorm. All types use $D = 128$ codes, allowing joint completion while retaining query type, ownership, and frame. Partial and complete branches share this map; masks and confidence condition completion separately. Unobserved codes are zeroed, and observed codes are denoted $\mathbf{z}^{\text{obs}}$.

Cross-component memory. Our generative completion module (GCM) combines a scene-conditioned flow transformer with constrained sampling (Fig. 3). We parameterize its completion velocity with a DiT-style transformer [26] trained by flow matching [18]. Its input combines current and observed codes, masks, confidence, type, both coordinate descriptions, cell scales, and pooled log features. Background groups follow spatial neighborhoods, each asset forms a local group, and sky directions are grouped by cubemap face. Local self-attention processes these groups separately. Cross-component memory (CCM) then exchanges context across groups:

$$\mathbf{M}' = \mathbf{M} + \text{Attn}_r(\mathbf{M}, \widetilde{\mathbf{X}}, \widetilde{\mathbf{X}}), \tag{6}$$

$$\mathbf{U} = \widetilde{\mathbf{X}} + \text{Attn}_w(\text{LN}(\widetilde{\mathbf{X}}), \mathbf{M}', \mathbf{M}'). \tag{7}$$

Here $\widetilde{\mathbf{X}}$ concatenates local tokens; attention arguments are queries, keys, and values. Memory first reads all local groups, then supplies the updated scene context to each group's tokens. A residual feed-forward update follows $\mathbf{U}$. Flow time modulates local normalization, and learned tokens plus log summaries initialize memory.

Flow training and constrained sampling. Let $\mathbf{z}_1$ be a complete target on the same query layout and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ be Gaussian noise. We interpolate only the unknown content at flow time $\tau \in [0, 1]$:

$$\mathbf{z}_\tau = m \odot \mathbf{z}^{\text{obs}} + u \odot [(1-\tau)\boldsymbol{\epsilon} + \tau \mathbf{z}_1]. \tag{8}$$

Observed codes remain fixed and free-space codes remain zero. The target velocity for unknown queries is $\mathbf{z}_1 - \boldsymbol{\epsilon}$. With an optional loss selector s_i, the objective is

$$\mathcal{L}_{\text{flow}} = \frac{\sum_i u_i s_i \|\mathbf{v}_\theta(\mathbf{z}_\tau, \tau, C)_i - (\mathbf{z}_{1,i} - \boldsymbol{\epsilon}_i)\|_2^2}{D \max(1, \sum_i u_i s_i)}. \tag{9}$$

Prior learning uses samples with valid targets at every unknown query; s_i selects loss terms within these targets (Sec. 3.5).

At inference, we initialize unknown codes with noise and take S Euler steps with $\Delta\tau = 1/S$:

$$\mathbf{z}^{j+1} = \Pi_{m,f}\left[\mathbf{z}^j + \Delta\tau\, u \odot \mathbf{v}_\theta(\mathbf{z}^j, j/S, C)\right]. \tag{10}$$

After every step, $\Pi_{m,f}$ restores observed codes and clears free-space codes. The seed fixes a completion, decoded once for reuse across states.

Type-specific Gaussian decoding. Background, rigid, and human Gaussian decoders independently map each query to $K = 4$ child Gaussians whose centers lie in its support cell. Human children inherit the query's skinning weights; rigid children share their asset's rigid transform. Opacity is multiplied by $1 - f_i$, so free queries cannot produce visible Gaussians. A separate sky head predicts cubemap colors within the observation-supported domain V_{sky}. Gaussian and appearance parameterizations are detailed in Appendix E of the supplementary material.

Because decoding is query-wise, freezing the codec and retaining observed codes preserves observed Gaussian parameters. Free-ray losses constrain the visibility and alpha-compositing effects of added Gaussians, measured in Sec. 4.5.

3.4. State-driven scene composition

At time step k, state s_k specifies the ego pose, active assets, rigid transforms, and human root and joint poses. Rigid transforms update canonical Gaussian centers and covariances. Human joint poses deform them through the stored skinning weights, so completed surfaces move with the corresponding body parts. Appendix C gives the transformations. The renderer composites the background, active assets, and valid sky:

$$o_k = \mathcal{R}(\text{Compose}(\mathcal{W}, s_k), \mathcal{C}_k). \tag{11}$$

The background and canonical assets remain fixed; new camera and actor states reuse the same completed content.

3.5. Training procedure

We train recovery and the codec before learning the completion prior and adapting the GCM through rendering. Complete targets use the partial input's query layout $\mathcal{Q}$, including its cell scales, component ownership, and human bindings. Complete meshes supply surface samples within each cell. Multiview rendering and the frozen visual encoder produce $\mathbf{h}_i^{\text{full}}$, which the shared type-specific encoder maps to $\mathbf{z}_{1,i} = E_{k_i}([\mathbf{h}_i^{\text{full}}; \mathbf{q}_i; \boldsymbol{\delta}_i])$. Surface-free cells receive zero features and opacity supervision.

After codec fitting, recovery and the codec remain frozen. Stage 3 learns from complete synthetic targets at every unknown query. Stage 4 updates only the GCM, mixing complete synthetic samples and the 740 real training logs at 1:3. The four stages use 287,360 updates in total; Appendix B gives target construction, network sizes, and training settings.

4. Experiments

Waymo experiments test reconstruction accuracy and off-trajectory rendering quality. Paired hidden-surface tests and actor edits examine joint completion and reuse of fixed assets. Main results use center-ray rule C; footprint rule F is a separate controlled comparison. Appendix A reports zero-shot transfer to nuScenes.

4.1. Experimental setup

Data and supervision. We divide Waymo's 798 training sequences into 740 training and 58 development sequences, evaluating on 202 validation sequences. Synthetic training uses 10,000 scenes, 80 maps, 10,000 vehicles, and 2,500 human identities. Source maps, vehicle models, and human identities are split before creating appearance/motion variants; real logs are separated by road segment, capture, and identifiable actors. Appendix B gives all split sizes and supervision.

Baselines. STORM, DGGT, and StreetForward [3, 45, 48] are retrained from random initialization on the same real logs, synthetic scenes, and complete assets. Each uses its

Figure 4. **Conditional dynamic reconstruction.** Rows show five examples; columns are ground truth, DGGT, STORM, StreetForward, and Ours.

native supervision; Ours also uses cell-empty and template-binding targets. A shared state adapter assigns ownership and canonicalizes predicted Gaussians for rigid and skeletal placement. Methods share context RGB, calibration, ego poses, vehicle boxes/root tracks, and target human joints; shape and binding come from context. Test-time weights are fixed. Appendix F reports training costs and adaptation details.

Sampling and metrics. Each test sequence supplies 200 frames, forming 2,020 disjoint 20-frame clips at 10 Hz. Three front cameras provide context at $\{0, 5, 10, 15\}$ [45]; the remaining 12 interpolation and four extrapolation frames give 96,960 targets. Extrapolation renders supplied states. We evaluate 240×160 images [14] using full/dynamic PSNR and SSIM, and depth AbsRel/δ_1 at valid LiDAR points. Depth is aligned per frame for evaluation; Appendix F also reports unaligned results. Frames, clips, then sequences are averaged with equal sequence weight.

4.2. Dynamic scene reconstruction

DriveCompose leads all six Waymo metrics (Table 1), exceeding DGGT by 1.154 and 2.340 dB in full-image and dynamic-region PSNR. Results aggregate interpolation and extrapolation at 3:1; Appendix F reports both groups. Figure 4 shows clearer actor silhouettes and street structure.

Table 1. **Conditional reconstruction on Waymo.** Interpolation and extrapolation are aggregated at 3:1; Appendix F reports both groups.

	Full image		Dynamic Region		Scale-aligned depth	
Method	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	AbsRel ↓	δ_1 ↑
STORM	26.380	0.826	22.100	0.624	0.178	0.741
DGGT	27.410	0.794	22.710	0.651	0.249	0.668
StreetForward	27.010	0.818	20.919	0.592	0.197	0.759
Ours	**28.564**	**0.901**	**25.050**	**0.862**	**0.095**	**0.877**

Figure 5. **Off-trajectory rendering with a 2 m leftward shift.** Columns show GT, STORM, DGGT, StreetForward, and Ours. GT is the original-trajectory image.

4.3. Off-trajectory rendering

We shift cameras laterally by ± 1, ± 2, and ± 4 m while fixing orientation, intrinsics, and timestamps [6, 51]. Gaussian renders are compared before diffusion refinement. Each direction contains 121,200 images and 6,060 20-frame videos at 10 Hz, referenced to recordings from the same clips, cameras, and timestamps. Left/right FID and FVD are computed separately, then averaged (Table 2; Appendix H).

DriveCompose achieves the lowest FID and FVD at all three Waymo offsets, improving both image and video distribution quality (Table 2). Figure 5 compares shifted renders against an original-trajectory reference.

Table 2. **Off-trajectory distribution quality.** FID/FVD are computed separately for left and right shifts, then averaged.

Method	Shift ± 1 m		Shift ± 2 m		Shift ± 4 m	
	FID ↓	FVD ↓	FID ↓	FVD ↓	FID ↓	FVD ↓
STORM	61.97	722.27	83.72	1029.36	132.57	1704.16
DGGT	73.99	984.00	93.36	1197.03	128.01	1661.70
StreetForward	84.30	878.95	103.31	1095.02	123.58	1578.63
Ours	**28.52**	**544.27**	**37.43**	**673.17**	**58.19**	**817.13**

4.4. State-driven asset reuse

Asset edits. We test fixed-asset reuse on twenty real scenes, each containing one vehicle and one human. A 1 m camera shift, 30° vehicle rotation, and human pose change reuse fixed completed assets. Nine training/generation-seed combinations yield 360 actor edits. The five-pixel structural-integrity criterion passes 334/360 (92.78%); 298/360 (82.78%) jointly pass structure, newly visible texture, appearance identity, and local geometry checks. Appendix I reports the four-of-five rating rule, failure breakdown, and seed statistics.

Observation preservation. Compared with the observed-only scene at fixed cameras and actor states, Ours preserves observed Gaussian parameters exactly, while input-view PSNR decreases by 0.029 dB and depth MAE increases by 2 mm. Previously observed human texture has canonical linear-RGB MAE 0.014. Appendix J reports parameter and rendering measurements.

Efficiency. On an RTX 4090 (FP16, batch 1, rule C), construction takes 2.66 ± 0.11 s, single-camera composition/rendering 7.8 ± 0.4 ms at 240×160, and peak memory 14.2 GB. Reusing the constructed assets amortizes the total cost over 1,000 outputs to 10.46 ms each. Appendix K specifies timing conditions.

4.5. Ablation studies

The w/o GCM control disables non-observed, non-sky Gaussians. The w/o CCM control uses isolated per-component memories with the same parameters and global log context, retraining Stages 3–4.

Completion components. Waymo ablations use one 29-timestamp window per test sequence, three cameras, nested $T = 2, 4, 8$ inputs, and seven shared targets (Appendix

G). Variants share recovery, codec, states, data, and optimization budgets. Training samples T uniformly; inference uses 24 Euler steps. We average three training and three generation seeds, retaining every output in the scores. Exchanging component context adds 0.627/0.934 dB full-image/dynamic PSNR (Table 3); training-seed SDs are 0.061 and 0.074 dB for Ours and w/o CCM. PSNR gains are 1.104, 0.738, and 0.627 dB at $T = 2, 4, 8$, with the largest benefit at the sparsest input setting (Appendix G). CCM adds 5.1% construction time (Appendix K).

Table 3. **Component ablations on Waymo.** $T = 8$, with shared recovery, codec, and target views. Full SDs are in Appendix G.

Variant	Full image			Dynamic regions	
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑
w/o GCM	27.137	0.867	0.183	23.431	0.811
w/o CCM	28.286	0.893	0.147	24.683	0.849
Ours	**28.913**	**0.911**	**0.129**	**25.617**	**0.873**

Hidden-surface completion. One hundred synthetic test scenes provide paired hidden-region evaluation at ± 2 m. We adapt scene-level latent generation [17] and cross-object latent coupling [40] to shared recovery, codec, query layout, inputs, supervision, and sampling. Three training and three generation seeds are averaged. Joint completion improves both appearance and surface recovery, with gains of 0.67 dB PSNR, 2.6 points coverage, and 3.6 points F-score over object coupling (Table 4). On real common support, RGB reprojection MAE falls from 0.028 to 0.019 versus w/o CCM while coverage rises from 86.9% to 92.4%. Appendix H gives the mechanism interfaces and geometry evaluation.

Table 4. **Hidden-region completion mechanisms.** Task-adapted controls on 100 synthetic scenes, rule C. Coverage (Cov.) and F-score use 2 cm.

Mechanism	PSNR↑	LPIPS↓	Cov.↑	F-score↑
w/o GCM	16.82	0.401	61.5%	51.7%
w/o CCM	21.56	0.236	89.7%	81.3%
Scene latent	22.24	0.221	91.8%	84.1%
Object coupling	22.51	0.212	92.4%	85.0%
Ours	**23.18**	**0.192**	**95.0%**	**88.6%**

Visibility-stratified analysis. On the 100 synthetic scenes, input-union surface coverage defines well-observed ($\geq$ 80%) and severely missing ($\leq$ 20%) groups. CCM adds 1.02/1.28/1.54 dB for severely missing background/vehicle/human regions, versus 0.04/0.04/0.03 dB for well-observed regions. Appendix H reports fixed-checkpoint read-scope interventions.

Completion after editing. We apply the edits in Sec. 4.4 to twenty identity-disjoint synthetic scenes, each containing one vehicle and one human. Joint completion gives the best PSNR, LPIPS, and surface coverage for newly visible surfaces (Table 5).

Table 5. **Newly visible asset surfaces after editing.** Completion ablations in paired synthetic tests; vehicle/human coverage thresholds are 2/1 cm.

Variant	Vehicle (2 cm)			Human (1 cm)		
	PSNR↑	LPIPS↓	Cov.↑	PSNR↑	LPIPS↓	Cov.↑
w/o GCM	17.80	0.362	63.4%	17.12	0.391	60.8%
w/o CCM	22.46	0.218	91.8%	21.38	0.247	88.6%
Ours	**24.18**	**0.176**	**96.1%**	**23.06**	**0.201**	**94.3%**

Free-space constraints. On 2,020,000 fixed rays, added-occlusion and strict free-ray violation rates are 0.18% and 0.07%; removing the free-ray loss raises them to 1.34% and 0.91% (Appendix J). With weights, layout, depth, and per-query noise fixed, F reduces occupied-cell mislabels from 1,374 to 19. Hidden-region coverage rises from 95.0% to 96.2%, PSNR from 23.18 to 23.36 dB, and construction time from 2.66 to 2.86 s. Appendix D separates cell errors, surface coverage, and ray violations.

5. Discussion

DriveCompose reconstructs finite regions under fixed illumination, using rigid vehicles and humans bound to a fixed skeleton and clothing volume. Depth, identity, and canonical-pose accuracy determine the observed anchors inherited by completion. Residual reconstruction artifacts occur near pedestrians and the parked pickup in Fig. 4. Editing failures concentrate on large elbow bends, leg self-occlusion, and loose-clothing boundaries. Relighting and clothing-geometry changes call for richer appearance and deformation models.

6. Conclusion

DriveCompose jointly completes world-space background and canonical actor latents, preserving component identity and motion bindings for state-driven reuse. Shared scene context improves hidden geometry and appearance over task-adapted scene-latent and object-coupling controls. Fixed Gaussian assets support camera shifts, vehicle rotation, and human pose edits. Waymo reconstruction, paired surface metrics, and multidimensional real-edit ratings establish completion quality and state-driven reuse.

LLM Usage Statement

OpenAI Codex assisted manuscript revision, translation, LaTeX editing, and consistency checks. OpenAI Image Gen was used for the conceptual scene and method illustrations in Figs. 1–3. These illustrations are identified as schematic. The authors supplied the reported numerical results and retain responsibility for the scientific content.

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